

Detection of AI-Generated Product Reviews

*ECL 545 - Natural Language Processing Project Report
Executive M.Tech. in Applied AI, VNIT Nagpur*

(Group 24)

Ganesh Prasad D M (MT24AAI161)

Nikhil G Bharadwaj (MT24AAC003)

Nandisha K S (MT24AAC032)

1. Problem Statement

AI-generated fake reviews are increasingly flooding e-commerce platforms, misleading customers and distorting product ratings. Existing detection systems face major limitations: most act as opaque black-box models without explanations, struggle against unseen LLM-generated reviews, and often rely on costly cloud infrastructure unsuitable for academic or independent use.

This project proposes a hybrid, explainable, and locally deployable AI review detection system. It combines fine-tuned transformers, stylometric analysis, and rule-based heuristics through a gradient-boosted classifier. The system provides faithful explanations for every verdict and runs entirely offline on a single consumer GPU using free, open-source tools.

2. Exploratory Data Analysis

The training corpus combines the public theArijitDas/Fake-Reviews-Dataset with 5,885 additional AI-generated reviews produced locally via four open-source LLMs: Llama 3.1 8B, Mistral 7B, Qwen 2.5 7B, and Phi-3 Mini. The full corpus contains 46,411 labeled reviews across four splits, with no missing values.

Splits and label balance:

Split	Rows	Human	AI	AI %
train	35,972	16,156	19,816	55.1%
validation	4,496	2,053	2,443	54.3%
test	4,497	2,023	2,474	55.0%
heldout	1,446	0	1,446	100.0%
Total	46,411	20,232	26,179	56.4%

The train, validation, and test splits are mildly skewed toward AI (~55%) - manageable without class weights. The heldout split is all AI by design, containing only Qwen 2.5 7B outputs withheld from training entirely. It enables rigorous generalization testing - measuring whether the model can detect AI from a generator it has never seen.

Review length:

Class	Mean	Median	Std Dev	Min	Max
Human	73.6	42.0	76.1	5	373
AI-generated	67.6	39.0	64.4	1	318

Both classes overlap heavily in length (mostly 20–100 words). AI samples are slightly shorter and lower-variance but cannot be classified by length alone - preventing shortcut learning.

Lexical and vocabulary statistics:

Metric	Human (mean)	AI (mean)	Difference
Type-token ratio (vocabulary diversity)	0.809	0.742	−0.067
Average word length	4.204	4.140	−0.064
Unique words per review	50.82	41.59	−9.23

Type-token ratio is the cleanest separator: human reviews use ~9% more diverse vocabulary. Average word length is nearly identical, ruling out a simple 'AI uses fancier words' shortcut.

Word and phrase usage differences:

AI reviews over-use marketing and superlative vocabulary (changer, exceptional, sleek, top-notch, thrilled) and formulaic bigrams (would recommend, highly recommend, recently purchased). Human reviews over-use casual functional words (here, etc, actually, maybe, though, isn't) and concrete, time-anchored bigrams (year old, well made, set up, buy again).

3. Model Architecture

The detector is a four-stage hybrid pipeline. Each stage produces an output that flows into the next; the final stage produces the verdict and its explanation.

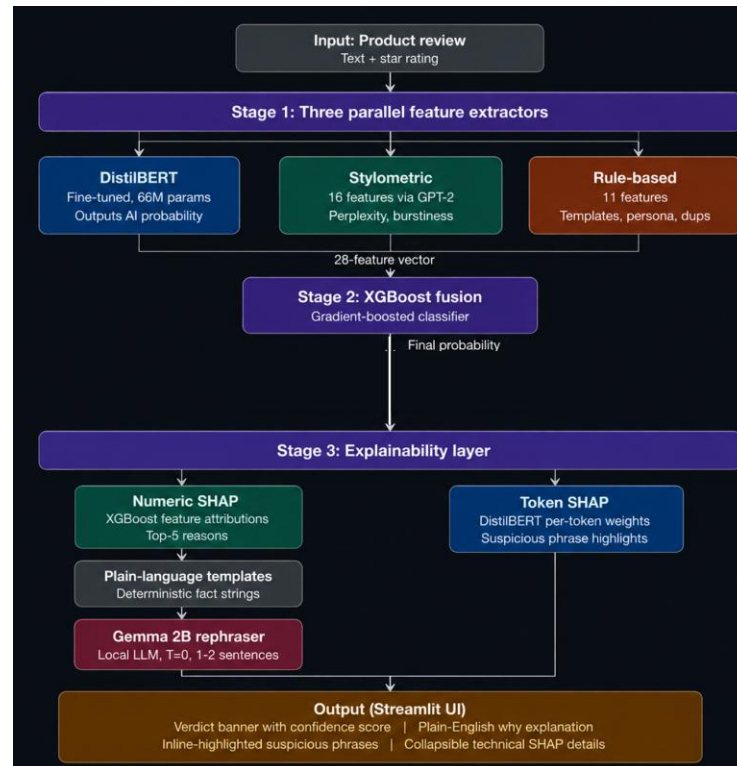


Fig 1. Architecture of the AI reviews detection system.

Stage 1 - Three parallel feature extractors.

All three see the same review text but extract fundamentally different evidence.

- DistilBERT: A fine-tuned transformer with 66M parameters. Outputs a single AI probability. Captures semantic and stylistic patterns learned from training data.
- Stylometric extractor: 16 numeric features measuring statistical text properties - perplexity (via GPT-2), burstiness, type-token ratio, sentence-length statistics, punctuation density, generic-adjective and hedging ratios. These features are model-agnostic; they generalize across LLM families.
- Rule-based extractor: 11 features capturing structural tells - known LLM template phrases, buyer-persona phrases, rating-style openers, markdown artifacts, Pros/Cons structure, emoji counts, and MinHash near-duplicate detection.

Stage 2 - XGBoost fusion.

The 28-dimensional feature vector (1 + 16 + 11) is fed to a gradient-boosted classifier (XGBoost, 400 trees, max depth 4). Trees naturally handle heterogeneous feature scales without normalization and learn non-linear feature interactions. The output is a single calibrated probability, thresholded at 0.49 to produce a likely_ai / uncertain / likely_human verdict.

Stage 3 - Explainability layer.

Two deterministic explanation sources run in parallel. Numeric SHAP attributions on the XGBoost model identify which features pushed the prediction toward AI; these are mapped through plain-language templates into human-readable reasons. Token-level SHAP on DistilBERT identifies which words in the review most contributed to the verdict, surfaced as inline red highlights.

Stage 4 - LLM rephrasing.

A small local LLM (Gemma 2B, running via Ollama at temperature zero) rewrites the technical reasons into one or two sentences of plain English. The LLM operates strictly downstream - it cannot change the verdict, re-rank reasons, or invent facts. The original technical reasons remain visible in a collapsible panel for verification.

4. Model Performance Results

DistilBERT training curve:

Epoch	Train Loss	Val Loss	Val Acc	Val Prec	Val Recall	Val F1	Val FPR
1	0.0972	0.2007	0.9242	0.8855	0.9881	0.9340	0.1520
2	0.0523	0.1004	0.9655	0.9490	0.9898	0.9689	0.0633
3	0.0254	0.1087	0.9686	0.9546	0.9894	0.9717	0.0560

Training loss halves each epoch, indicating healthy convergence. Validation FPR drops sharply from 15.2% to 5.6% across the three epochs. Mild overfitting begins at epoch 3 (val loss rises slightly) but F1 continues to improve, so the epoch-3 checkpoint is retained as the best model.

DistilBERT test-set performance:

Set	Accuracy	Precision	Recall	F1	FPR
Validation	0.9686	0.9546	0.9894	0.9717	0.0560
Test (in-distribution)	0.9706	0.9574	0.9907	0.9738	0.0539
Held-out generator (Qwen 2.5)	0.9550	1.0000	0.9550	0.9770	0.0000

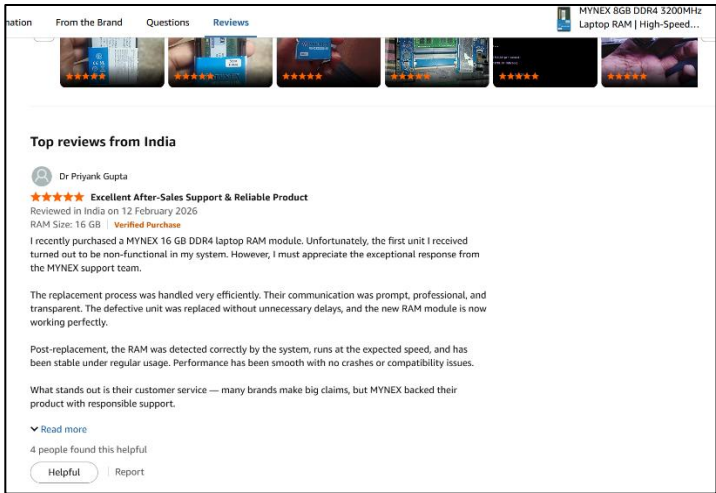
Test performance closely matches validation, ruling out overfitting. On the held-out generator (Qwen 2.5, never seen during training), recall is 95.5% - only 3.5 points below in-distribution recall. This proves the model learned generalizable AI-detection features rather than memorizing specific generators.

Hybrid fusion performance:

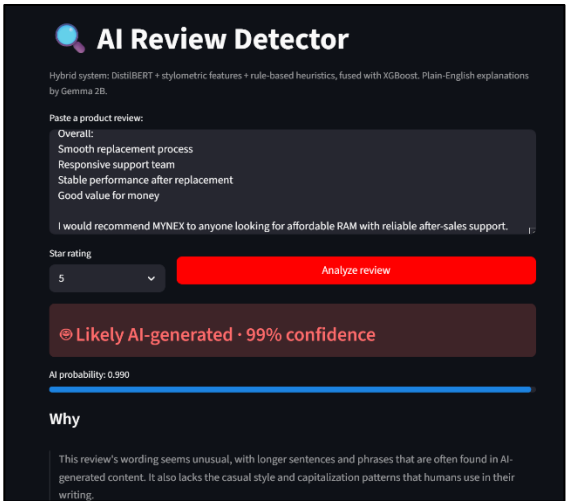
Configuration	Accuracy	Precision	Recall	F1	FPR
DistilBERT alone	0.9700	0.9570	0.9899	0.9732	0.0544
Fusion (default 0.5)	0.9722	0.9730	0.9766	0.9748	0.0331
Fusion (tuned 0.49)	0.9726	0.9730	0.9774	0.9752	0.0331

Adding stylometric and rule-based features to DistilBERT reduces the false positive rate from 5.44% to 3.31% - a 39% relative reduction. Recall drops only 1.25 percentage points. For a user-facing product where false positives destroy trust, this is a clearly favorable trade-off.

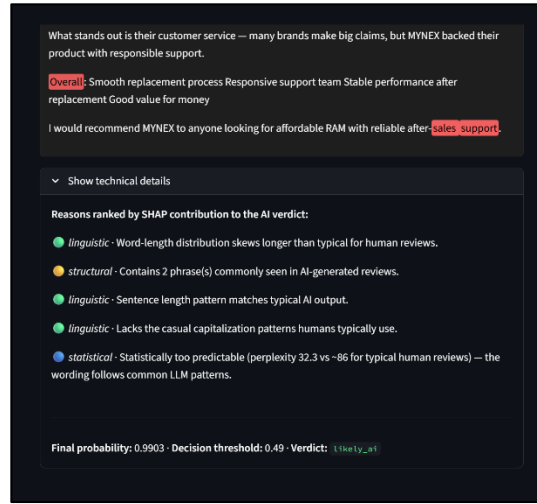
5. Snapshot of demo



(a)

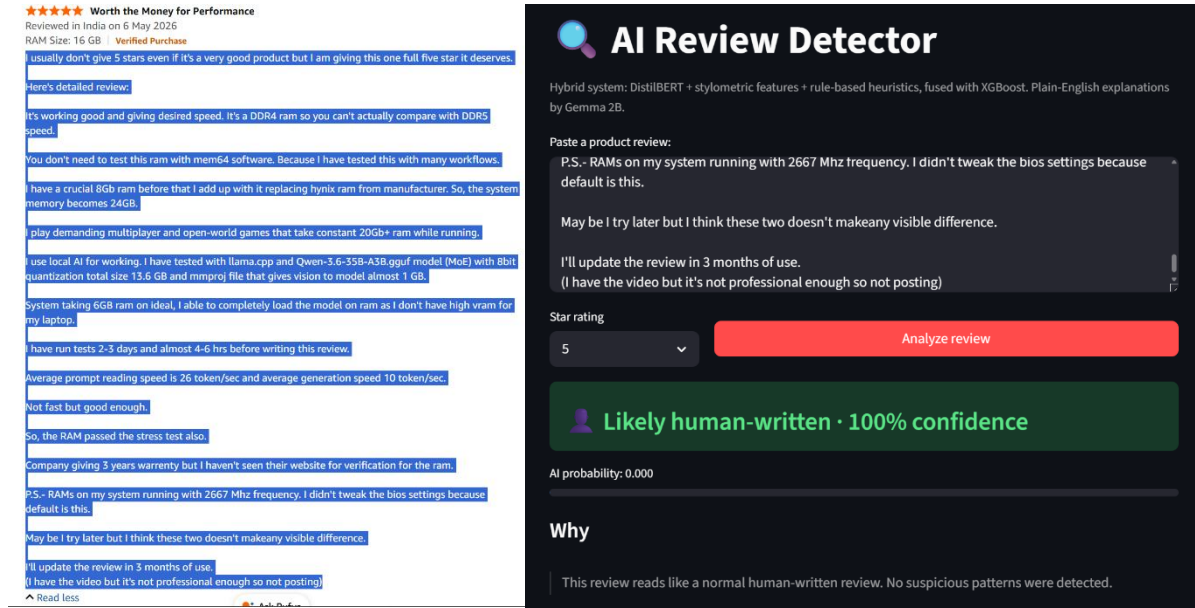


(b)



(c)

Fig 2. (a) Customer review found on an E-commerce website. (b) The system detected the review being written using AI, (c) and its explanatory part of why the system considered it AI.



(a)

(b)

Fig 3. (a) Customer review found on an E-commerce website. (b) (b) The system detected the review being written by human.

6. Conclusion

This project demonstrates a working AI review detector that addresses the three primary shortcomings of existing approaches: opacity, generator-specificity, and infrastructure dependence. The hybrid architecture - DistilBERT, stylometric features, and rule-based heuristics fused through XGBoost - achieves 97.26% test accuracy, an F1-score of 0.9752, and a false positive rate of 3.31%, representing a 39% reduction over DistilBERT alone. On a held-out generator (Qwen 2.5 7B) never seen during training, the model retains 95.5% recall, demonstrating genuine cross-generator generalization.

The explainability layer combines deterministic SHAP attributions with template-based fact strings and a small local LLM that rephrases technical findings into plain English. The LLM operates strictly downstream of the detection logic - it cannot affect verdicts and cannot invent reasons - preserving the system's reproducibility and trustworthiness. Every claim shown to the user maps back to a verifiable computation.

The complete pipeline runs offline on a single consumer GPU and is presented through a local Streamlit interface. No paid APIs, no proprietary services, and no web scraping are involved at any stage of training, inference, or explanation. The system is therefore reproducible, trustworthy, and deployable in environments with no cloud budget - meeting the original project goals in full.